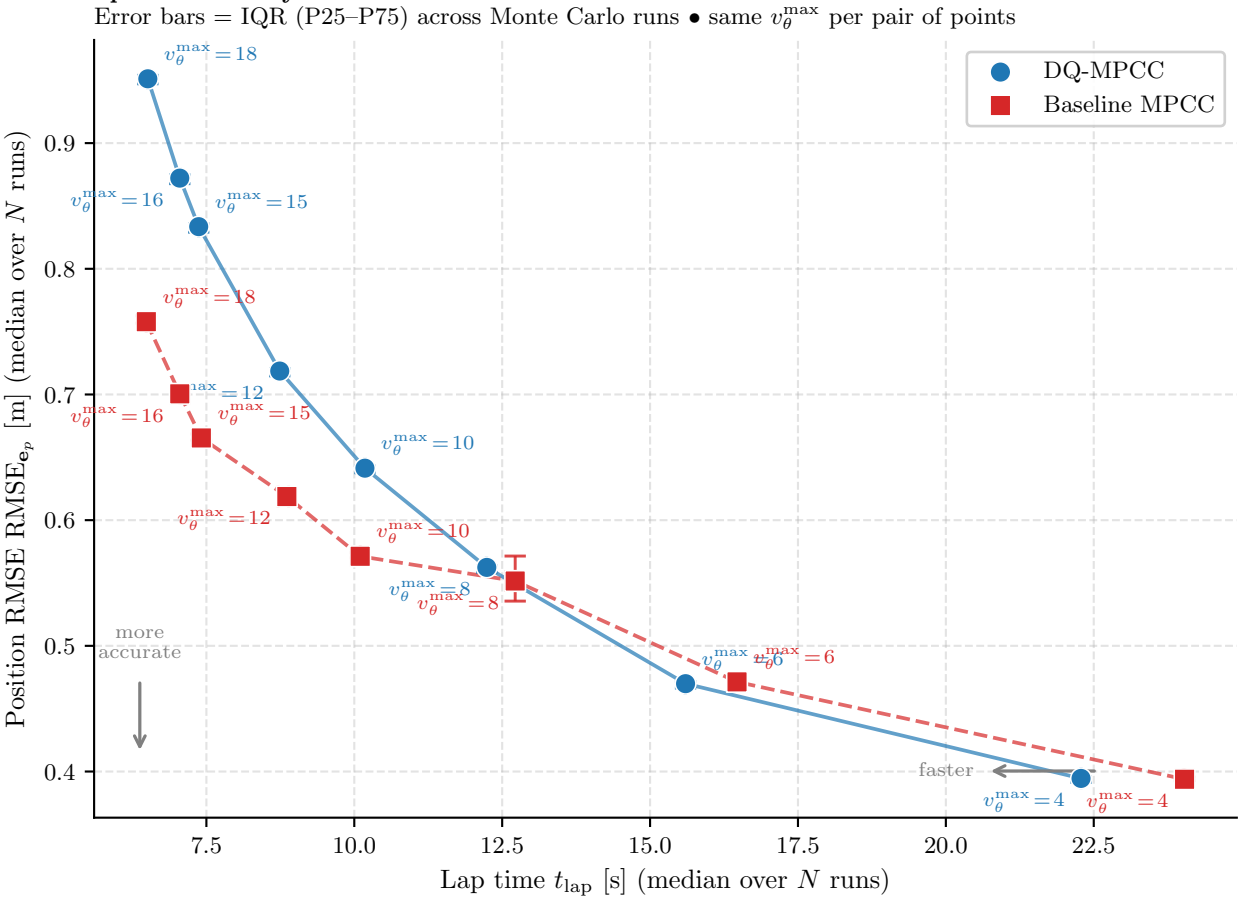
Speed–Accuracy Pareto Frontier
Error bars = IQR (P25–P75) across Monte Carlo runs • same $v_\theta^{\max}$ per pair of points
DQ-MPCC
Baseline MPCC
$v_\theta^{\max} = 18$
$v_\theta^{\max} = 16$
$v_\theta^{\max} = 15$
$v_\theta^{\max} = 10$
$v_\theta^{\max} = 8$
$v_\theta^{\max} = 6$
$v_\theta^{\max} = 4$
$v_\theta^{\max} = 18$
$v_\theta^{\max} = 16$
$v_\theta^{\max} = 15$
$v_\theta^{\max} = 12$
$v_\theta^{\max} = 10$
$v_\theta^{\max} = 8$
$v_\theta^{\max} = 6$
$v_\theta^{\max} = 4$
more accurate
faster
Position RMSE $\mathrm{RMSE}_{\mathbf{e}_p}$ [m] (median over N runs)
Lap time t_{lap} [s] (median over N runs)
0.4
0.5
0.6
0.7
0.8
0.9
7.5
10.0
12.5
15.0
17.5
20.0
22.5